The Underserved Have Less Access to Employer-Sponsored Telemedicine Coverage

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Abstract

Telemedicine has been proposed as one means of improving health care access for underserved communities, and information about insurance coverage for telemedicine (TMC) is important in understanding its utilization and provision. We use 2018-2019 Medical Expenditure Panel Survey-Insurance Component data on employer-sponsored coverage to examine pre-pandemic TMC relative to employer, worker, and health plan characteristics. We find that the share of employees in private sector establishments offering TMC was lower in the most rural counties, in smaller firms, in establishments without unionized employees, and in establishments where most workers were low wage, part-time and older when compared to other establishments. These findings reflect differences across establishments in insurance offers, as well as differences in TMC conditional on an insurance offer, which suggests that TMC may function as a premium plan feature with limited availability and potential support for improving healthcare access for the underserved.

JEL Classification: I13, J32

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Introduction

For decades, the provision of clinical services via telecommunications (i.e., telemedicine), has been proposed as one means of improving health care access, especially for underserved communities.[1] The number of primary care telemedicine visits covered by commercial insurance increased by roughly 6.0 percent, from 1.5 million in 2018 to 1.6 million in 2019 (Alexander et al., 2020), and the COVID-19 pandemic and related 2020 social distancing restrictions led to temporary rule changes including those by the Centers for Medicare and Medicaid in Medicare's telemedicine reimbursement, provision, and eligibility that further spurred utilization. However, differences in social determinants of health, access to health care in underserved rural communities, and the lack of broadband for more than 30 million Americans continue to perpetuate long standing disparities in health care access (Weiner et al., 2021; White House, 2021; Department of Health and Human Services, 2020). Recent legislative initiatives to address these disparities include proposals to increase broadband access under President Biden's recent American Jobs Plan.

In the years leading up to the pandemic, reimbursement policies were cited as a major barrier to telemedicine's adoption (Lee, Karsten, and Roberts, 2020; Center for Connected Health Policy, 2019). In 2019, only sixteen states had passed private payer laws that required payment parity, while all but fourteen states had passed coverage parity laws, with mixed findings on their impact (American Telemedicine Association, 2019; Park et al., 2018; Mehrotra et al., 2017).[2] Over the years, more states have passed legislation on private payer reimbursement, and between 2015 and 2019, the percentage of large firms offering health benefits with telemedicine coverage (TMC) dramatically increased from 27 percent of firms in 2015 to 82 percent in 2019 (Rae, Cox, and Glaxton, 2020).

Increasing our knowledge about access to TMC under insurance plans is important to understanding how coverage relates to telemedicine utilization and provision. This is particularly relevant for underserved communities where there is a substantial risk that these locations, which might benefit most from telemedicine as a technology, could also be among those communities facing more limited TMC. This paper builds upon the existing literature and provides a pre-COVID pandemic baseline for showing employee access to TMC under employer-sponsored insurance plans (ESI) and will allow future comparison with subsequent trends during and after the pandemic. This research helps to answer the question of whether these historical TMC patterns on average have reinforced, or instead worked against, long standing disparities in health care access.

We address these research questions by using pooled data from the 2018 and 2019 Medical Expenditure Panel Survey-Insurance Component (MEPS-IC) to examine the shares of private sector and state and local government employees working in establishments that offer at least one health insurance plan with TMC. To provide more information on the sources of differences in access to TMC, we estimate the share of employees working in establishments that offer any health insurance plan, and the share of employees with access to TMC conditional on an offer of insurance.[3]

Results show that the share of employees in establishments offering TMC was lower in the most rural counties, in single-unit and smaller firms, in establishments with no unionized employees, and in establishments where a majority of workers were low-wage, part-time and 50 years old or older than in other establishments. These results reflect long-standing differences across establishments in offers of any health insurance plans, but in most cases differences in access to TMC are further increased by differences in access to TMC conditional on an offer of insurance. Differences in conditional offers suggest that TMC may have functioned as a premium plan feature which is less likely to be made available to traditionally underserved groups of employees, thereby limiting telemedicine's potential for improving healthcare access.

This paper begins by describing the data used to estimate TMC for employees by worker and employer characteristics and for enrollees in plans with and without TMC. The following sections present our findings and discuss how these results relate to the current literature, including the implications for understanding existing disparities in access to TMC in the nation's primary group insurance market.

Data and Methods

We examine TMC under ESI plans using data first collected in the 2018 and again in the 2019 Medical Expenditure Panel Survey-Insurance Component (MEPS-IC). The survey provides respondents with the following definition, "Telemedicine is the delivery of health care through telecommunications to a patient from a provider who is at a remote location." The MEPS-IC, the nation's foremost data on ESI offers and coverage, is a nationally

[3] The share with access to a health plan with TMC equals the percent with any ESI offer multiplied by the percent with access to TMC conditional on an offer of ESI.

representative survey of establishments fielded annually by the U.S. Census Bureau under sponsorship by the Agency for Healthcare Research and Quality.[4]

To increase precision, the data are pooled for 2018 and 2019 and are used to produce unconditional and conditional estimates of the share of workers with access to TMC. The unconditional estimates are the percentage of all employees that work in an establishment that offers at least one plan with TMC. The conditional estimates provide the same information among establishments that offered at least one health insurance plan. All estimates are weighted by the establishment's number of employees. It is important to note, however, that not all employees are eligible for offered coverage.

The unconditional estimates provide a comprehensive look at access to TMC among the employees of a given set of establishments, with the employment shares having access to TMC naturally being upper bounded by the share that have access to ESI. The conditional estimates, meanwhile, reflect the employer characteristics associated with offering ESI and help to characterize their decisions about features to include in their health plans, conditional on offering insurance.

Exhibits 1 and 2 present estimates separately for private sector and state and local government employees, respectively, and show both ESI and TMC offerings for select employer and worker characteristics.[5] The statistical tests presented in the Exhibits are simple tests of differences in means, implemented via univariate regressions with complex survey adjustments to tailor them to the MEPS-IC environment. These tests assess differences in access to TMC between employees of establishments with a given characteristic (e.g., retail trade) and the remaining sample (e.g., all other industries). We use approximate t-tests to assess differences in means that are not shown in the Exhibits. Unless otherwise noted, all differences discussed in the text are significant at $p<0.05$, or better.

We place each county into one of three urbanicity categories which are constructed by collapsing the nine Rural-Urban Continuum Codes (RUCCs) produced by U.S. Department of Agriculture's Economic Research Ser-

[4] Despite its name, the insurance component of the MEPS is not a panel but a repeated cross-sectional survey of establishments. The MEPS-IC collects data from private sector employers and state and local governments. For more information on the MEPS-IC, visit Medical Expenditure Panel Survey Insurance/Employer Component Overview (ahrq.gov).

[5] MEPS-IC data collected from state and local governments do not support estimates for the percent fifty years of age or older and the percent women.

vice for 2013 using the Census Bureau's 2010 Decennial Census. The three county categories are metropolitan, non-metropolitan adjacent to metropolitan, and nonmetropolitan and not adjacent to a metropolitan area to designate the most rural geographies.

Exhibit 3 presents both private and government plan-level estimates to compare premiums, deductibles and other features for plans that do, and do not, provide TMC, and we again use simple tests for differences in means for significance checks.

Results

Private Sector Employees

Pooled estimates for 2018-2019 show that about three-quarters (74.1 percent) of all private sector employees were in establishments that offered at least one ESI plan with TMC (Exhibit 1). This occurred as 85.0 percent of all employees were in establishments that offered at least one ESI plan, and 87.2 percent of these employees were in establishments that offered TMC conditional on offering at least one ESI plan.

A key potential benefit of TMC is increasing access to care in underserved rural areas. However, estimates in Exhibit 1 show that private sector employees in the most rural counties (i.e., non-metropolitan, non-adjacent counties) were much less likely than their counterparts in metropolitan counties to work in establishments that offered an ESI plan with TMC (62.2 versus 75.3 percent). This deficit reflects both the lower percentage of employees in rural counties that work in establishments that offered any ESI plans (75.9 percent versus 85.9 percent), but it also reflects the lower percentage of rural employees in establishments that provide TMC conditional on an offer of insurance (82.0 percent versus 87.7 percent).

Figure 1 provides more information on access to TMC in rural areas by illustrating the association of the state-level percentage of private sector employees in establishments with TMC and the percentage of states' populations that live in a rural area.[6] In four states - Maine, Mississippi, Arkansas, and Montana – a relatively large share of the population (greater than 40 percent) lives in a rural county, but a relatively low share of private sector

[6] In Figure 1, state-level employment shares are based on the private sector establishment's physical location, whereas the state's rural population is based on residential addresses. The estimates do not account for workers who may live and work in different states or workers from professional employer organizations (PEOs).

workers (less than 65 percent) work in establishments that offer a plan with TMC. By contrast, seven states – Washington, California, Nevada, Arizona, Texas, Massachusetts and Connecticut – have relatively small rural populations (less than 20 percent), and a relatively high share of workers (greater than 75 percent) with access to TMC.

Similar to longstanding patterns of employee access to ESI offers, the largest differences in the availability of TMC are observed for small versus large firms, and single- versus multi-unit firms. For example, only 18 percent of private sector employees in the smallest firms (1-9 Employees) worked in establishments that offered ESI plans with TMC compared to 92.6 percent of employees in the largest firms (1,000 or more Employees). The overall deficit in access to TMC for employees in the smallest firms reflects large differences in the percentage of employees in the establishments that offered any ESI plans (30.1 percent versus 99.6 percent) as well as large differences in establishments that provide TMC, conditional on an offer of insurance (59.6 percent versus 92.9 percent). More generally, the percentages of employees with unconditional and conditional access to TMC increased monotonically as firm size increased and were much larger in multi-unit firms than in single-unit firms.

Exhibit 1 also shows that the share of employees in establishments offering TMC was lower in establishments with no unionized employees, and in establishments where a majority of workers were low-wage, part-time and 50 years old or older than in other establishments. For example, 65.0 percent of employees in establishments with a majority of low-wage workers have access to TMC compared with 76.8 percent in other establishments.

State and Local Government Employees

Overall, state and local government employees (82.5 percent) were more likely than private sector employees (74.1) to work in establishments that offered at least one ESI plan with TMC (Exhibit 2). This occurred as public sector employees were much more likely than private sector employees to work in establishments that offered at least one ESI plan (99.3 versus 85.0 percent) but were less likely to have access to TMC conditional on an ESI offer (85.9 percent versus 87.2 percent).

In contrast to the private sector, there is not a large difference in access to TMC for rural and urban public sector employees. Overall, the percent of state and local government employees in establishments that offered at least one ESI plan is significantly higher in metropolitan counties (99.7 percent) than in the most rural counties (96.8

percent); however, conditional on this offer, access to TMC is higher for rural government workers (88.0 percent) than those working in metropolitan counties (82.5 percent). As a result, the difference between the percentages of rural (85.1 percent) and urban (82.2 percent) public sector employees is not statistically significant. State-level results shown in Online Resource 1 also suggest a greater level of access to TMC in the public sector than in the private sector. For example, more than half of West Virginia's population is considered rural (51 percent), but almost all state and local government employees (99.0 percent) work in establishments that offer at least one health plan with TMC. Alaska with about one third of its population considered rural, presents a stark difference between sectors with 68.0 percent of private establishment employees with access to TMC compared to 96.7 percent of employees from state and local governments.

Similar to the private sector, the percentages of employees working in establishments that offer TMC increases monotonically with the size of the government unit, with the notable exception that smaller percentages of state and local government employees from the largest units (1,000 or more employees) work in establishments that offer TMC (80.6 percent) compared to units with 50-99 employees (88.1 percent) and 100-999 employees (90.4 percent). In addition, the percentage of government workers in the largest units with access to TMC is smaller than the percent in the largest private sector firms (80.6 percent versus 92.6 percent).[7] Similar to our findings for private sector employees, those working for government units with unionized workers are more likely to have access to TMC (un)conditional on an ESI offer.

Employer-Sponsored Health Plan Characteristics

Exhibit 3 presents characteristics of plans with and without TMC and shows that larger shares of private sector enrollees with TMC (60.8 percent) are in self-insured plans compared to enrollees without TMC (49.4 percent). This is not surprising given that larger firms are more likely to self-insure and to offer an ESI plan with TMC. In contrast, smaller shares of enrollees with TMC (19.3 percent) are in exclusive provider network plans compared to those enrollees without TMC (26.5 percent). For private sector enrollees, if a family plan provides TMC, the average total premium is $290 higher than for plans without TMC, but no significant difference is seen for single premiums. However, deductibles paid out-of-pocket by individuals and families are lower for enrollees with TMC, although the

[7] The public sector employs a larger share of the country's unionized workers than the private sector. Multi-year union contracts and negotiated health plan benefits may explain this finding if government units are unable to make coverage changes (e.g., for telemedicine) as frequently as the private sector.

$101 difference in average family deductibles between plans with and without TMC is only weakly significant ($p<0.10$). Single coverage enrollees with TMC have an average deductible that is $63 less (3 percent lower) than plans without TMC.

In contrast to the private sector, smaller shares of state and local government enrollees with TMC are in self-insured plans (66.7 percent) compared to those without TMC (73.3 percent), which is almost 24 percentage points higher than for private sector enrollees (49.4 percent). However, similar to private sector plans, the average premium for family coverage plans with TMC ($20,940) is higher than for plans without TMC ($20,460) with a $480 difference ($p<0.10$). But unlike private sector enrollees, deductibles for family coverage for state and local government enrollees with TMC are $243 higher than for plans without TMC.

Discussion

This research helps to answer the question of whether historical patterns of access to TMC have reinforced, or instead worked against, long standing disparities in health care access. A key potential benefit of TMC, in this regard, is increasing access to care in underserved rural areas. Our results, however, show that private sector employees in the most rural counties were much less likely than their counterparts in metropolitan counties to work in establishments that offered an ESI plan with TMC. By contrast, the share of public-sector employees with access to TMC was higher overall than in the private sector, and there were no statistically significant differences in access across rural and urban areas. Results also show that in both the private and public sectors, the share of employees in establishments offering TMC was lower in smaller firms (or government units), and in establishments with no unionized employees. The latter result is consistent with unionization being associated with more generous health benefits (Buchmueller, DiNardo, and Valletta, 2002). Private sector employees were also less likely to have access to TMC in establishments where a majority of workers were low-wage, part-time, 50 years of age or older, or women ($p<0.10$) than in other establishments.

Taken together, our results suggest that differences in access to TMC by sector, the degree of urbanicity, and worker characteristics tend to work against employees that traditionally have lower levels of access to care. For example, rural areas face barriers to health care access due to a greater likelihood of workforce shortages and less stable provider networks than urban areas (Department of Health and Human Services, 2020). In addition, women

and low-wage earners in the private sector may face greater challenges in accessing health care since both groups, as shown in Exhibit 1, are less likely to work in an establishment that offers ESI. Their access is likely further compromised by the fact women are overrepresented among low-wage workers, most low-wage workers are in their prime working years (25 to 50 years old), and these workers are most likely to be raising children (Long and Marquis, 2001; Vistnes and Selden, 2011; Bishaw and Posey, 2016; Ross and Bateman, 2019).

Differences across groups in access to TMC largely reflect longstanding disparities in the percentages of employees working in establishments that offer any ESI plans; but in many cases, access to TMC conditional on an ESI offer further accentuates these patterns. That is, employees who are more likely to work in an establishment that offers any ESI plan, also have higher rates of conditional access to TMC. This is most clearly illustrated by firm-size differences across the private sector where the percentages of employees in establishments that offer an ESI plan, and that offer TMC conditional on an ESI offer both monotonically increase as firm size increases. As a result, the percentage of employees with access to TMC is more than five times greater in the largest firms (92.6 percent) than in the smallest firms (18.0 percent). In spite of the high rate of access, findings from the 2021 Large Employers' Health Care Strategy and Plan Design Survey conducted by the Business Group on Health on 122 large employers that cover more than 9 million employees and their dependents suggest that large employers plan to continue expansion of TMC.

The pattern of differences across groups in conditional offers of TMC raises the question of whether TMC may have functioned as a premium plan feature and would, therefore, be more likely to be included in plans with other desirable features. Evidence on this is mixed. In both the private and public sectors, we find that plans that include TMC are less likely to have exclusive provider networks, so TMC enrollees tend to have a wider choice of providers. Private sector plans with TMC are also more likely to be self-insured – which allows greater flexibility in plan design – but public sector plans with TMC are less likely to be self-insured. Small differences in premiums and deductibles exist, but do not provide strong, consistent distinctions across plans with and without TMC.

Systematic reviews of the efficacy and utilization of telemedicine interventions conclude that they are generally similar to in-person care, but caution that many factors need to be considered including "modalities, evidence quality, population demographics, and outcome measurement" (Shigekawa et al., 2018). This paper describes private and public sector employee access to TMC under ESI plans in the two years prior to the COVID-19 pandemic and before temporary policy changes in 2020 led to its expanded use and provision. Increases in the

prevalence of state-level private payer reimbursement policies may help to further incentivize greater ESI coverage of telemedicine. In addition, the Department of Health and Human Services announced in 2021 that the Biden-Harris administration is investing over $19 million to help strengthen telemedicine services and to improve access, including in rural and underserved areas. Continued analyses of telemedicine's utilization, provision, efficacy, and cost effectiveness will be needed to evaluate its full potential for improving our health care system. Our future analyses will compare 2018-2019 TMC estimates with those that measure access during and after the pandemic to continue evaluating the potential impact of TMC on healthcare access for the underserved.

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Exhibit 1 Private sector employee access to telemedicine coverage

	Percent of Employees (2018 and 2019)		
	Offered Plan with Telemedicine	*Offered Plan with Telemedicine Conditional on Insurance Offer*	*Offered Any Plan*
Total	74.13***	87.24***	84.98***
Year			
2018	72.79***	86.01***	84.63+
2019	75.47***	88.45***	85.32+
Urbanicity			
Metropolitan County	75.33***	87.68***	85.91***
Metropolitan Adjacent County	65.90***	84.15***	78.32***
Not Metropolitan Adjacent County	62.20***	82.00***	75.86***
Establishment Characteristics			
For-Profit	73.08***	87.53+	83.48***
Not-for-Profit	80.16***	85.77+	93.46***
Single-unit firm	50.60***	78.21***	64.70***
Multi-unit firm	89.95***	91.22***	98.60***
Majority Low-wage Workers	64.95***	88.12	73.71***
Non-majority Low-wage Workers	76.75***	87.03	88.19***
Majority Part-time Employees	58.04***	85.55*	67.85***
Majority Full-time Employees	77.86***	87.54*	88.94***
Has Union Employees	89.95***	90.89***	98.08***
No Union Employees	71.12***	86.37***	82.35***
Majority Female Employees	73.61+	87.74+	83.89***
Majority Male Employees	74.72+	86.69+	86.19***
Majority Over 50 Years of Age	63.84***	83.60***	76.36***
Majority Under 50 Years of Age	75.86***	87.78***	86.42***
Firm Size			
1-9 Employees	17.96***	59.62***	30.13***
10-49 Employees	46.36***	74.56***	62.18***
50-99 Employees	74.10	82.81***	89.49***
100-999 Employees	83.99***	86.17+	97.47***
1000+ Employees	92.57***	92.92***	99.62***
Industry			
Agriculture/Fishing/Forestry	61.42***	82.64	74.32***
Mining/Manufacturing	82.41***	87.47	94.22***
Construction	62.46***	81.51***	76.63***
Utilities/Transportation	80.59***	88.41	91.15***
Wholesale Trade	77.15*	85.55	90.17***
Financial Services/Real Estate	83.69***	90.27***	92.71***
Retail Trade	75.80+	89.43***	84.75
Professional Services	76.85***	86.21*	89.15***
Other Services	65.27***	87.39	74.68***

Authors' analysis using Medical Expenditure Panel Survey-Insurance Component. Estimates using survey weights multiplied by establishment employment. Statistical significance assessed via simple means tests on each insurance offering outcome within the subsample of interest (i.e., all establishments or all establishments making an insurance offer) on each employer characteristic versus the rest of the sample. +, *, **, *** denote significance at the 0.1, 0.05, 0.01, and .001 levels, respectively.

Exhibit 2 Characteristics of State and Local Government Employers Offering Telemedicine Coverage (2018 - 2019)

	Percent of Employees		
	Offered Plan with Telemedicine Coverage	*Offered Plan with Telemedicine Coverage Conditional on Insurance Offer*	*Offered Any Plan*
Total	82.49***	83.03***	99.34***
Year			
2018	85.35+	85.93+	99.33
2019	79.63+	80.15+	99.36
Urbanicity			
Metropolitan County	82.24	82.49*	99.70***
Metropolitan Adjacent County	83.80	86.79	96.55***
Not Metropolitan Adjacent County	85.12	87.96*	96.78***
Establishment Characteristics			
Majority Low-wage Workers	84.36	87.63	96.27***
Non-majority Low-wage Workers	82.42	82.87	99.45***
Majority Part-time Employees	85.25	90.09***	94.63***
Majority Full-time Employees	82.21	82.37***	99.81***
Has Union Employees	86.44***	86.75*	99.65***
No Union Employees	76.76***	77.62*	98.90***
Size of Government Unit			
1-9 Employees	41.53***	76.54	54.25***
10-49 Employees	67.97***	85.48	79.51***
50-99 Employees	88.12*	88.87*	99.15
100-999 Employees	90.43***	90.67***	99.74***
1000+ Employees	80.55***	80.55***	100.00***

Authors' analysis using the Medical Expenditure Panel Survey-Insurance Component. Estimates using survey weights multiplied by establishment employment. Statistical significance assessed via simple means tests on each insurance offering outcome within the subsample of interest (i.e., all establishments or all establishments making an insurance offer) on each employer characteristic versus the rest of the sample. +, *, **, *** denote significance at the 0.1, 0.05, 0.01, and .001 levels, respectively.

Exhibit 3 Employer-Sponsored Health Plans with Telemedicine Coverage (2018 - 2019)

Health Plan Characteristics	*Plans Covering Telemedicine*	*Plans Not Covering Telemedicine*	*Any Plan*
Private Sector Plans			
Self-Insured (% enrollees)	60.82***	49.40***	58.60
Exclusive Provider Network (% enrollees)	19.29***	26.45***	20.68
Single Coverage Plans			
Average Premium ($)	6,923	6,944	6,927
Average Deductible ($)	1,742*	1,805*	1,754
Family Coverage Plans			
Average Premium ($)	19,890*	19,600*	19,840
Average Deductible ($)	3,682+	3,783+	3,701
Public Sector Plans			
Self-Insured (% enrollees)	66.65***	73.30***	68.21
Exclusive Provider Network (% enrollees)	30.85*	38.08*	32.54
Single Coverage Plans			
Average Premium ($)	8,232	8,228	8,231
Average Deductible ($)	1,083	1,025	1,070
Family Coverage Plans			
Average Premium ($)	20,940+	20,460+	20,820
Average Deductible ($)	2,367*	2,124*	2,312

Authors' analyses using the Medical Expenditure Panel Survey-Insurance Component. Plan enrollment weighted means of the row variable within the sample specified by the columns. Statistical significance is assessed using a simple test of means of each plan characteristics between plans with and without telemedicine coverage. +, *, *** are statistically significant at the 0.1, 0.05, and 0.001 levels respectively.